

SLAM project – P0 desk work summary

RF person-localization with two 900 MHz panel antennas on a Keysight E5071C · 23 Sep 2026

1. What was done

A new **slam/** Python package was added to the repo (github.com/jasonlle/SLAM), alongside the legacy folders which were left untouched. All new code was written against a shared interface spec (**docs/CONTRACT.md**) by five parallel agents, then a sixth agent installed everything in a fresh environment, ran the whole chain end-to-end, fixed seven bugs and wrote **docs/REVIEW.md**. Total: ~3,000 lines of code and documentation, no lab time needed.

File	Purpose	Status
requirements.txt	Pinned deps (pyvisa + pyvisa-py, numpy, pandas, scipy, sklearn, matplotlib, opencv, termcolor); installs on Py 3.10–3.13, Windows wheels verified	installs clean
slam/config.py	Single source of truth: VNA address, 902–928 MHz, 201 pts, 1 kHz IF BW, avg 4, S11/S21/S22; taped grid A1..E5 (1 m); antenna/room geometry (placeholders marked TODO_MEASURE); CSV + manifest column definitions	done
slam/vna.py	VNA class: connect, configure full 2-port sweep, sweep() -> Sweep(freq, {S11,S21,S22} complex), measure_sweep_time; MockVNA with identical interface for offline work; CSV read/write	done – SCPI untested on real ENA
slam/collect.py	Position-labelled collector: --session/--label/--person/--n, or interactive 'C3 10' prompt loop; countdown, Ctrl-C safe; writes _.csv, manifest.csv, session_info.json	done, tested with mock
slam/analyze_sensitivity.py	The go/no-go tool for the first lab session: empty-room reference, complex difference per S-param, detectability ratio = $\text{mean} dS / \text{empty RMS}$, 4 figures + summary.csv + GO/NO-GO verdict	done
slam/sim/digital_twin.py	Phase-A digital twin (numpy image method + point-scatterer person, panel pattern, blockage, S11 backscatter, drift, noise). Emits sessions in the exact collector format; --set overrides any parameter; 150 sweeps in <1 s	done
docs/digital_twin.md	Research write-up: simulator options (Sionna RT, MATLAB, Wireless InSite, HFSS/CST, FDTD), required fidelity, human phantom/RCS/shadowing values from literature, validation plan, 3-phase plan, paper figures	done, URLs checked
docs/REVIEW.md	Reviewer report: what ran, bugs fixed, hardware items to verify, run cheat-sheet	done

2. Verified end-to-end (offline)

Mock collector → analysis, simulator → analysis, and the built-in fixture all run headless. On the simulated session below (10 sweeps per label, seed 1) the analysis reports **S21 clearly detectable at 5/6 positions**, S11 informative only near antenna 1 and at the near corner, S22 the mirror image near antenna 2 — exactly the physics we expected: S21 carries the middle of the square, each antenna's reflection says 'near me or not'. The real session must reproduce this pattern before the fall plan is on solid ground.

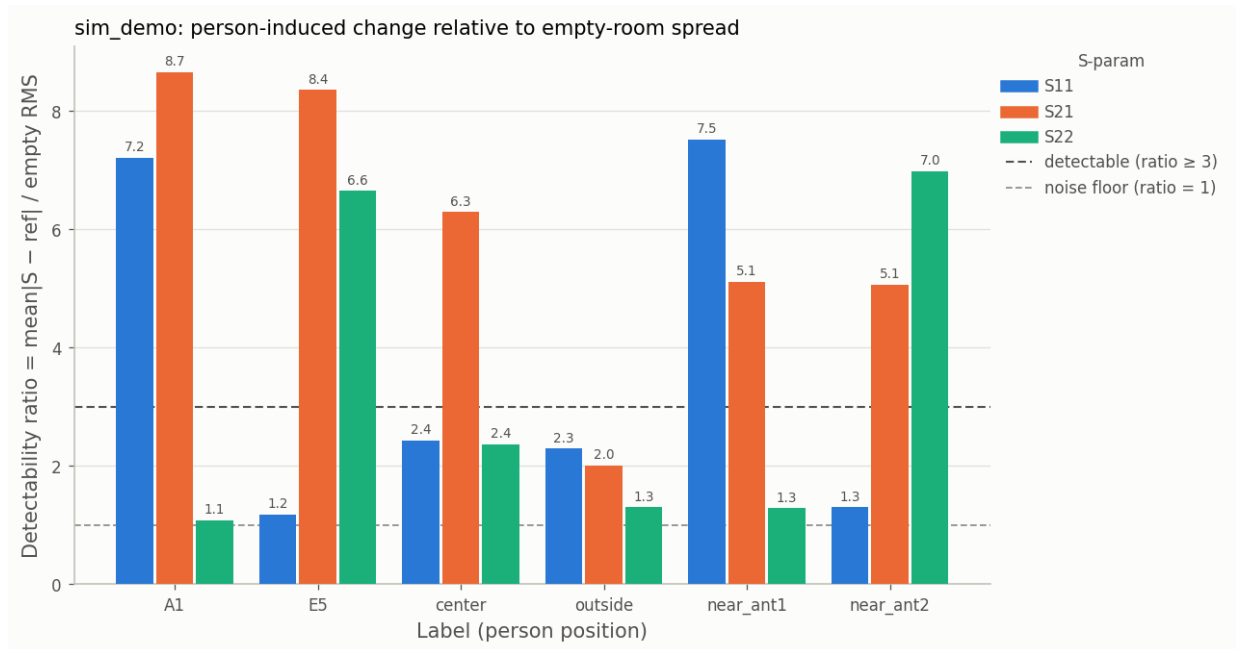


Figure: detectability ratio per label and S-parameter on simulated data (dashed line at 3 = clearly detectable).

Bugs caught by the review (all fixed)

- Simulator drift default made an *empty* room look S11-detectable after ~8 min — a conclusion-changing error; defaults lowered and documented.
- collect.py opened the VNA twice (leaked VISA session on real hardware); connect() made idempotent.
- Mock seed made every collector run bit-identical; label case 'c3' vs 'C3' split one position into two; CSV loader crashed on empty cells; 29-label figures unreadable — all fixed.

3. How to run

```
cd SLAM
python -m venv .venv && .venv/bin/pip install -r requirements.txt # (Windows: .venv\Scripts\pip)
python -m slam.config # print settings; fill in TODO_MEASURE values first
python -m slam.vna # real ENA: IDN, settings, sweep time (--mock offline)
python -m slam.collect --session 2026-09-27_sens --label empty --person jason --n 20
python -m slam.collect --session 2026-09-27_sens --label center --person jason --n 10
python -m slam.collect --session 2026-09-27_sens --person jason # interactive: "A1 10", "E5 10", ...
python -m slam.analyze_sensitivity --session 2026-09-27_sens # figures/<session>/*.png + summary.csv
python -m slam.sim.digital_twin --session sim_all --labels all --n 5 --seed 0
python -m slam.analyze_sensitivity --session sim_all
```

4. What needs to happen next

Before the first lab session (desk)

- Measure and enter into config.py: antenna x/y/z and tilt relative to the grid origin, antenna model from the L-com label, cable lengths, room dimensions, where the taped corner sits in the room. Everything is snapshotted into session_info.json, so do this **before** collecting.
- Tape and label the interior grid points (A1..E5, 1 m spacing) and mark 'near_ant1/2' at 0.5 m along the diagonal from each antenna corner.
- Commit the new package to GitHub (nothing has been committed yet; git status shows docs/, requirements.txt, slam/ as new).

First lab session with the new rig (P0 close-out, target this week)

- Run **python -m slam.vna** on the real ENA. The SCPI sequence (TRIG:AVER, SENS1:BWID, CALC1:PARn:DEF, INIT1:IMM + *OPC?, CALC1:DATA:SDAT?) is copied from the working legacy scripts plus averaging/IF-BW commands that could not be tested here — REVIEW.md lists each one. If configure() raises on SYST:ERR?, read the message; the likely culprit is TRIG:AVER on old firmware.
- Record the real sweep time at 201 pts / 1 kHz IF BW / avg 4 — it sets the live-demo update rate.
- Lock the rig: tape tripod feet, counterweight the off-axis panel, strain-relieve coax, confirm both panels have the same polarization, park the chair outside the square.
- Sensitivity run: 20 empty sweeps over ~10 min, then ~10 sweeps each at center, A1, E5, outside, near_ant1, near_ant2 (interleave a few more 'empty' sweeps between labels to see drift). Run analyze_sensitivity. **Decision:** if S21 ratio ≥ 3 at most positions, keep the geometry; if not, lower one or both antennas and repeat. Write the outcome into the roadmap.

P1 (Sep 28 – Oct 18)

- Dataset v1 campaign with collect.py on the full grid (2 people, 2 days, ~3–4k sweeps).
- features.py (dB magnitude + unwrapped phase, differenced against the per-session empty reference) and the zone classifier with leave-one-session-out evaluation — never a random split.
- Digital twin Phase B: calibrate system_loss_db, reflection coefficients and RCS so simulated empty-room level/ripple and the center/corner deltas match Dataset v1; report the sim-to-real gap. Then use the twin for the antenna-placement what-ifs (e.g. antennas at 1.6 m) that the paper's ablation table needs.
- Optional: build the Sionna RT scene of the lab (docs/digital_twin.md, Phase B) for domain-randomised pretraining.

Open risks

- With antennas at ~2.7 m the direct ray passes above a person; the twin predicts the person mainly blocks the floor bounce, giving a 2–4 dB S21 change. Real change may be smaller — hence the sensitivity test before anything else.
- Day-to-day VNA/room drift is unknown; the per-session empty reference is the mitigation, but interleaved empty sweeps are needed to quantify it.
- All simulator conclusions are Phase-A (first-order reflections, no cabinets, point-scatterer person) until calibrated against real data.